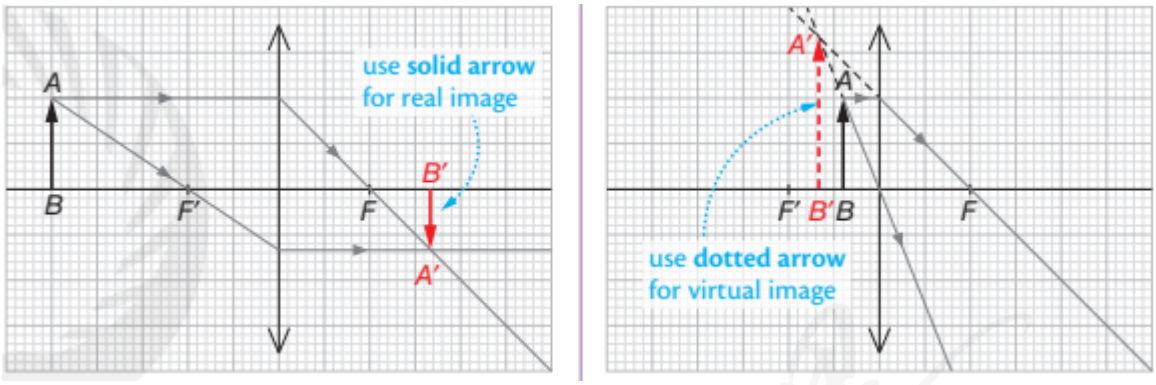
1. Real and virtual images

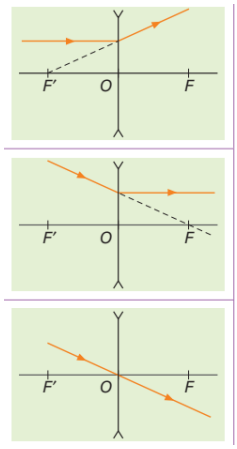
A real image is formed by actual convergence of rays and can be captured on a screen. A virtual image is formed where backward ray extensions meet. The rays do not pass through it, so it cannot be directly projected. Virtual images are still visible because light entering the eye appears to originate from their positions.



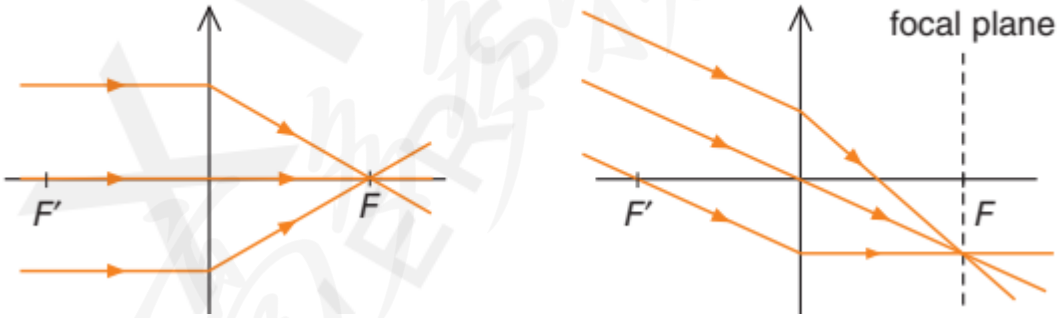
Left: an object beyond the focus forms an inverted real image where rays converge. Right: an object within the focal distance forms an upright virtual image where backward extensions meet.

2. Concave-lens rays and the focal plane

Use three rays to construct a concave-lens image. A ray parallel to the axis emerges as if from the near focus. A ray aimed at the far focus emerges parallel to the axis. A ray through the optical centre continues straight. Oblique parallel rays entering a convex lens converge in the focal plane, possibly away from the principal focus.



Three construction rays for a concave lens, showing the axis, focal points and ray directions.



Oblique parallel rays converge in the focal plane after passing through a convex lens.

3. Image formation by a convex lens

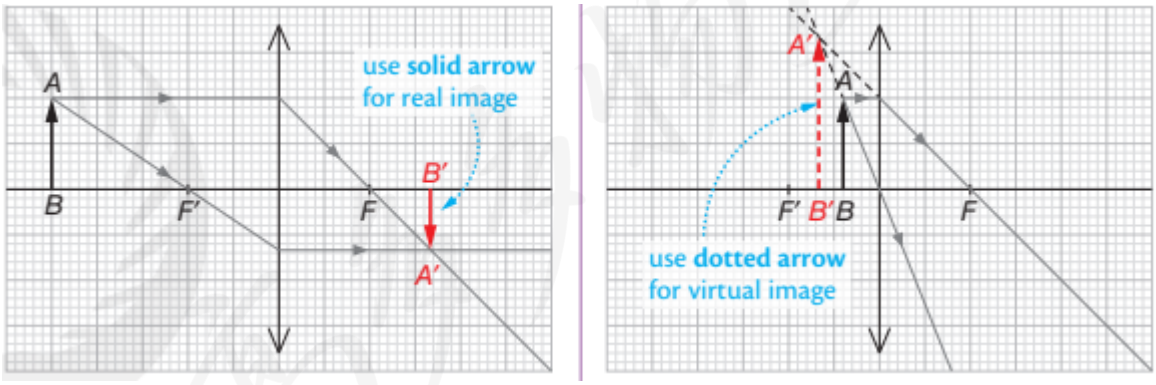
For a real object in front of a convex lens, the image depends on object distance. Beyond the focal distance, the image is real and inverted. Inside the focal distance, it is virtual, upright and enlarged. At the focal point, emerging rays are parallel and no finite-distance image forms. Real-image size also varies: diminished beyond twice the focal length, equal-sized at twice it, and enlarged between one and two focal lengths.

4. Linear magnification

Linear magnification compares image height with object height and can also be obtained from the magnitudes of image distance and real-object distance. Values above 1 mean enlargement and below 1 mean reduction. This card uses a non-negative size ratio, so orientation is stated separately.

$$m = \frac{h_i}{h_o} = \frac{|\,v\,|}{u}$$

Linear magnification as a size ratio is non-negative. h_i, h_o: image and object heights (same units); u > 0: real-object distance; v: signed image distance. The textbook's v/u uses distance magnitudes here; orientation is stated separately.



Left: an object beyond the focus forms an inverted real image where rays converge. Right: an object within the focal distance forms an upright virtual image where backward extensions meet.

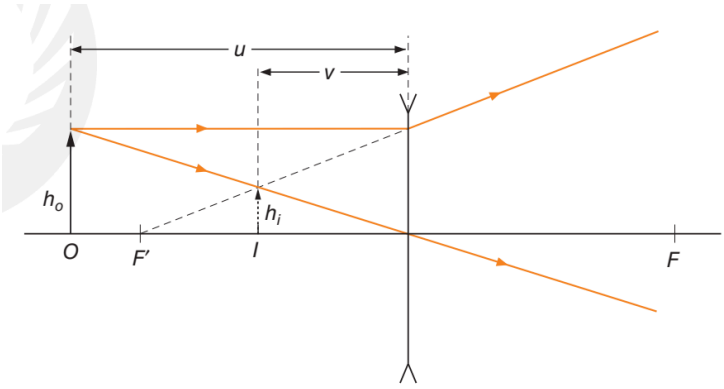


Fig 3.3e *Linear magnification of the image formed by a concave lens.*

Diverging rays trace back to a virtual, upright, smaller image on the object side. The diagram labels distance magnitudes; signed lens calculations use v < 0.

5. Images formed by a concave lens

A concave lens diverges incident light. For a real object in front, backward extensions of emerging rays meet on the object's side, forming an upright, diminished virtual image between the lens and its near focal point. Use a ray parallel to the principal axis and a ray through the optical centre to locate it.

6. The thin-lens equation and signs

The thin-lens equation relates object distance, image distance and focal length for paraxial rays. This page takes real-object and real-image distances as positive, and virtual-image distance as negative. Focal length is positive for a convex lens and negative for a concave lens. Decide the image type and signs, then use consistent length units before substitution. For an object at infinity, reciprocal object distance is zero.

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$$

Thin lens, paraxial rays. Real object: u > 0; real image: v > 0; virtual image: v < 0. Converging lens: f > 0; diverging lens: f < 0. All distances in the same units.

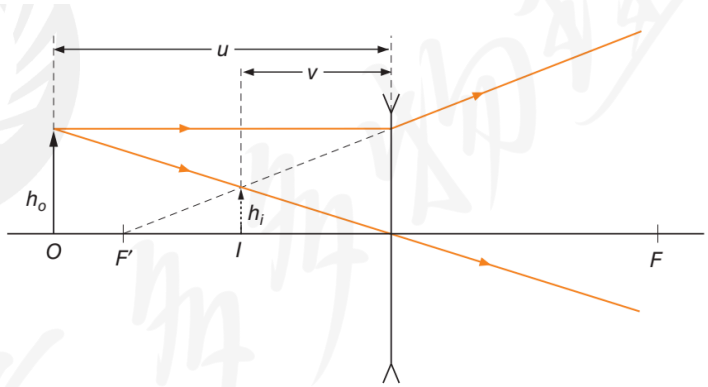
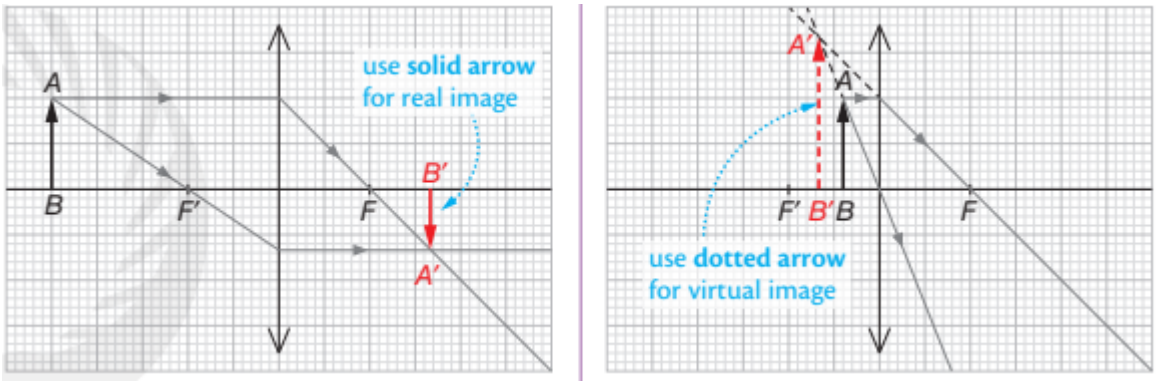


Fig 3.3e *Linear magnification of the image formed by a concave lens.*

Diverging rays trace back to a virtual, upright, smaller image on the object side. The diagram labels distance magnitudes; signed lens calculations use v < 0.



Left: an object beyond the focus forms an inverted real image where rays converge. Right: an object within the focal distance forms an upright virtual image where backward extensions meet.